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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

C. Lightning McQueen

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given an **undirected unweighted** graph with N nodes and M edges. The nodes are numbered from 1 to N. The graph contains no self-loops or multiple edges.

There is a source and a destination. Your task is to find the shortest distance from the source node to destination node and print the path taken. If multiple shortest paths exist, print the one that is lexicographically smallest.

A path $P1 = [a_1, a_2, \ldots a_n]$ is lexicographically smaller than a path $P2 = [b_1, b_2, \ldots b_m]$ if at the first position where they differ, $a_i < b_i$. For example, $[1, 4, 3]$ is smaller than $[1, 5, 7, 1]$.

If no path exists, print -1.

Input

The first line contains four integers N, M, S, D $(1 \leq N \leq 2 \times 10^5, 0 \leq M \leq 3 \times 10^5, 1 \leq S, D \leq N)$ — the number of vertices, total number of edges, source and destination.

The second line contains M integers $u_1, u_2, u_3 \ldots u_m$ $(1 \leq u_i \leq N)$ — where the i-th integer represents the node that is one endpoint of the i-th edge.

The third line contains M integers $v_1, v_2, v_3 \ldots v_m$ $(1 \leq v_i \leq N)$ — where the i-th integer represents the node that is other endpoint of the i-th edge.

The i-th edge of this graph is between the i-th node in the second line and the i-th node in the third line.

Output

- If a path exists, print the length of the shortest path (number of edges) on the first line.
- On the second line, print the lexicographically smallest shortest path from source to destination.
- If no path exists, print -1.

Examples

input Copy

```
5 10 5 3
2 1 5 3 1 4 2 4 1 4
5 5 4 5 2 2 3 1 3 3
```

output Copy

```
1
5 3
```

input Copy

```
5 4 2 5
5 1 2 4
1 3 3 2
```

output Copy

CSE221 Assignment 05 Summer 2026
Contest is running
4 weeks
Contestant

→ Submit?
Language: Java 21 64bit
Choose file: Choose File No file chosen
Success!
CLOUDFLARE
Privacy • Help
Submit

```
3
2 3 1 5
```

input

```
8 7 3 2
7 7 3 2 2 8 5
2 6 7 4 1 4 1
```

output

```
2
3 7 2
```

input

```
6 6 6 5
1 2 1 5 5 3
5 1 4 2 4 2
```

output

```
-1
```

input

```
1 0 1 1
```

output

```
0
1
```

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The only programming contests Web 2.0 platform
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